

CRANBROOK
SCHOOL

Term 3, 2009

Year 12 Mathematics Extension 1

Trial HSC Examination

Tuesday August 4th, 2009

Time Allowed: 2 hours, plus 5 minutes reading time

Total Marks: 84

There are 7 questions, all of equal value.

Submit your work in seven 4 Page booklets.

All necessary working should be shown in every question.

Full marks may not be awarded if work is careless or badly arranged.

Board of Studies approved calculators may be used.

A list of standard integrals is attached to the back of this paper.

Question 1 (12 marks) Use a separate writing booklet

Marks

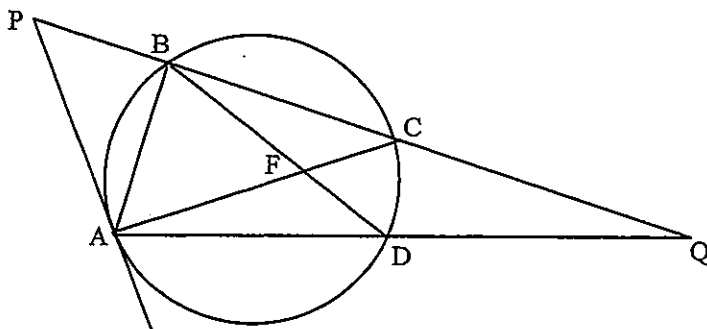
- (a) Evaluate: $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$ **1**
- (b) Find: $\frac{d}{dx} \left[\ln \sqrt{\frac{1+x}{1-x}} \right]$ **2**
- (c) Evaluate: $\int_{-3}^3 \frac{dx}{x^2 + 9}$ **2**
- (d) State the domain and range of the function : $f(x) = 2 \cos^{-1} 3x$ **2**
- (e) The variable point $(2\cos\theta, 3\sin\theta)$ lies on a curve. Find the cartesian equation of this curve. **2**
- (f) Use the substitution $\sqrt{x} = u$ to evaluate: $\int_1^4 \frac{dx}{x + \sqrt{x}}$ **3**

Question 2 (12 marks) Use a separate writing booklet

(a) Solve: $3^{x+1} = 5$. Give your answer correct to two decimal places. 2

(b) Solve: $x^3 + 2x^2 - 5x - 6 = 0$ 2

(c) NOT TO SCALE



In the above figure, AP is a tangent to the circle at A. PBCQ and ADQ are straight lines. Prove that $\angle PAB = \frac{1}{2}(\angle CFD + \angle CQD)$ 3

(d) Evaluate: $2 \int_0^{\frac{\pi}{4}} \cos^2 4x \, dx$ 3

(e) Find the general solution to: $\cos 5\theta - \cos 2\theta = 0$ 2

Question 3 (12 marks) Use a separate writing booklet

Marks

- (a) If the domain of $y = x^2 - 4x$ is restricted to a monotonic increasing curve:
- (i) sketch $y = f(x)$ 1
 - (ii) find the inverse function $y = f^{-1}(x)$ 2
 - (iii) state the domain and range of the inverse function 1
- (b) (i) Show that $f(x) = 3 \sin 2x - x$ has a root between 1.33 and 1.34. 1
- (ii) Starting with $x = 1.33$, use one application of Newton's method to find a better approximation for this root correct to 4 decimal places. 3
- (c) Consider the graph of $y = \frac{x^2}{4 - x^2}$
- (i) Write down the asymptotes of the function. 1
 - (ii) Find any stationary points and determine their nature. 2
 - (iii) Sketch the graph. 1

Question 4 (12 marks) Use a separate writing booklet**Marks**

- (a) (i) Prove by mathematical induction

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2 \quad \text{for } n \geq 1 \quad 4$$

- (ii) Hence evaluate: $2^3 + 4^3 + 6^3 + \dots + 20^3$ 1

- (b) The polynomial $P(x) = 2x^3 - 5x^2 - 3x + 1$ has zeros α , β and γ . Find the values of

(i) $3\alpha + 3\beta + 3\gamma - 4\alpha\beta\gamma$ 2

(ii) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ 1

(iii) $\alpha^2 + \beta^2 + \gamma^2$ 1

- (c) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The chord PQ subtends a right angle at the origin. If $pq = -4$ prove that the locus of the midpoint of PQ is a parabola with vertex $(0, 4a)$. 3

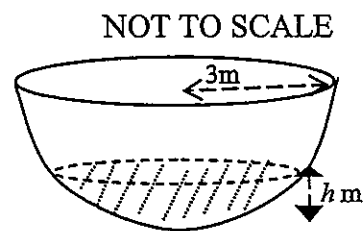
- (a) The interval AB is divided internally in the ratio 5:6 at the point R. If A and B have co-ordinates (2, 3) and (5, 4) respectively find the co-ordinates of R.

2

- (b) (i) Show that 1 is a root of $h^3 - 9h^2 + 8 = 0$ and find the other roots. 2

- (ii) A hemi-spherical bowl has a radius of 3m. Oil is poured into the container at a constant rate of $\pi/3$ m³/min. When the depth is h metres, the volume of oil is

$$V = \frac{\pi}{3}(9h^2 - h^3) \text{ m}^3.$$



- (α) How deep is the oil after 8 minutes?

2

- (β) At what rate is h increasing at this time?

2

- (c) The acceleration of a π -meson moving in a straight line is given by:

$$\ddot{x} = \frac{-4}{(x+2)^2}, \text{ where } x \text{ is the displacement in metres from a fixed point O.}$$

Initially the π -meson is 1 metre to the left of O and travelling with a velocity of 6 ms^{-1} in $\rightarrow$. Find the velocity of the π -meson when it is 6m to the right of O.

4

Question 6 (12 marks) Use a separate writing booklet**Marks**

- (a) The increase and decrease of pollution readings, x , in the skyline of Mexico City may be taken as simple harmonic according to the equation $\ddot{x} = -n^2(x - b)$, where $x = b$ is the centre of motion. A high pollution reading of 45 parts per million occurs at 6am on a particular day and a low pollution reading of 5 parts per million occurs at 11.30am on the same day.
- (i) Prove that $x = b + a \cos nt$ satisfies $\ddot{x} = -n^2(x - b)$. 2
- (ii) Find the earliest time interval on this day after 6am that a Mexican pigeon trainer Ms Swinivia Flutos can release her pigeons into the atmosphere for training if the pigeons cannot tolerate pollution readings of more than 15 parts per million. 5
- (b) (i) Using the result for $\tan(A+B)$, prove that
- $$\tan(A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A} \quad 2$$
- (ii) Given A, B and C are angles of a triangle and
- $$\frac{\tan A}{5} = \frac{\tan B}{6} = \frac{\tan C}{7} = k, \text{ show that } k = \sqrt{\frac{3}{35}} \quad 2$$
- (iii) Hence calculate the smallest of the angles to the nearest minute. 1

Question 7 (12 marks) Use a separate writing booklet**Marks**

- (a) A cup of soup with a temperature 95°C is placed in a room which has a temperature of 20°C . In 10 minutes the cup of soup cools to 70°C . Assuming the rate of heat loss is proportional to the excess of its temperature above room temperature, that is

$$\frac{dT}{dt} = -k(T - 20),$$

- (i) show that $T = 20 + Ae^{-kt}$ is a solution of

$$\frac{dT}{dt} = -k(T - 20). \quad 1$$

- (ii) find the temperature of the soup after a further 5 min. to the nearest degree. 2

- (iii) how long will it take the soup to cool to 35°C ?
Give your answer correct to the nearest minute. 1

- (iv) find the rate of cooling when the soup is 35°C .
Give your answer correct to 1 decimal place. 1

- (b) If $\sin x - 7 \cos x = -5$, $0 \leq x \leq 2\pi$, find x correct to 2 decimal places. 3

- (c) Sketch $y = \tan^{-1}(\sin 3x)$, $0 \leq x \leq \pi$, by firstly finding the existence of any stationary points and determining their nature. 4

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE : $\ln x = \log_e x$, $x > 0$

$$(a) \lim_{x \rightarrow 0} \frac{\sin 3x}{2x} = \left(\lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \right) \cdot \frac{3}{2}$$

$$= 1 \cdot \frac{3}{2}$$

$$= \frac{3}{2} \quad \checkmark$$

$$(b) \frac{d}{dx} \left[\ln \sqrt{\frac{1+x}{1-x}} \right]$$

$$= \frac{d}{dx} \left[\frac{1}{2} (\ln(1+x) - \ln(1-x)) \right]$$

$$= \frac{1}{2} \left[\frac{1}{1+x} + \frac{1}{1-x} \right] \quad \checkmark$$

$$= \frac{1}{2} \left[\frac{1-x+1+x}{1-x^2} \right]$$

$$= \frac{1}{1-x^2} \quad \checkmark$$

$$(c) I = \int_{-3}^3 \frac{dx}{x^2+9}$$

$$= \frac{1}{3} \left[\tan^{-1} \frac{x}{3} \right]_{-3}^3 \quad \checkmark$$

$$= \frac{1}{3} \left[\tan^{-1} 1 - \tan^{-1}(-1) \right]$$

$$= \frac{1}{3} \left[\frac{\pi}{4} + \frac{\pi}{4} \right]$$

$$= \frac{\pi}{6} \quad \checkmark$$

$$(d) f(x) = 2 \cos^{-1} 3x$$

$$\text{Domain is: } -1 \leq 3x \leq 1$$

$$\therefore -\frac{1}{3} \leq x \leq \frac{1}{3} \quad \checkmark$$

$$\text{Range is: } 0 \leq y \leq 2\pi \quad \checkmark$$

$$(e) x = 2 \cos \theta \quad \therefore \cos \theta = \frac{x}{2}$$

$$y = 3 \sin \theta \quad \sin \theta = \frac{y}{3}$$

$$\text{But } \sin^2 \theta + \cos^2 \theta = 1 \quad \checkmark$$

$$\therefore \left(\frac{y}{3} \right)^2 + \left(\frac{x}{2} \right)^2 = 1$$

$$\therefore \frac{x^2}{4} + \frac{y^2}{9} = 1 \quad \checkmark$$

the Cartesian equation of this curve.

$$(f) \text{ Let } \sqrt{x} = u \quad \therefore \frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$$

$$\text{when } x=1 \quad u=1 \quad \therefore du = \frac{1}{2\sqrt{x}} dx$$

$$x=4 \quad u=2 \quad \therefore 2u du = dx$$

$$\therefore I = \int_1^2 \frac{2u du}{u^2+u} \quad \checkmark$$

$$= 2 \int_1^2 \frac{du}{u+1} \quad \checkmark$$

$$= 2 \left[\ln(u+1) \right]_1^2 \quad \checkmark$$

$$= 2 \left[\ln 3 - \ln 2 \right]$$

$$= 2 \ln\left(\frac{3}{2}\right) \quad \checkmark$$

$$2.(a) 3^{x+1} = 5$$

$$\therefore \ln(3^{x+1}) = \ln 5$$

$$\therefore (x+1) \ln 3 = \ln 5 \quad \checkmark$$

$$\therefore x+1 = \frac{\ln 5}{\ln 3}$$

$$\therefore x = \frac{\ln 5}{\ln 3} - 1$$

$$\therefore x = 0.46 \quad (2 \text{ d.p.}) \quad \checkmark$$

$$(b) x^3 + 2x^2 - 5x - 6 = 0$$

$$\text{Let } P(x) = x^3 + 2x^2 - 5x - 6$$

$$\text{Possible zeros are: } \pm 6, \pm 3, \pm 2, \pm 1.$$

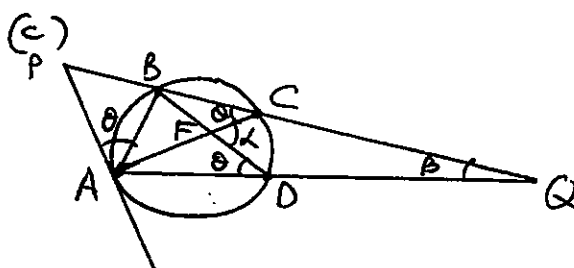
$$\text{Let } x = -1 \quad \therefore P(-1) = -1 + 2 - 5 - 6 = 0 \quad \checkmark$$

$$\therefore x+1 \text{ is a factor}$$

$$\therefore P(x) = (x+1)(x^2+x-6)$$

$$= (x+1)(x+3)(x-2)$$

$$\therefore \text{If } P(x) = 0 \quad \therefore x = -1, -3 \text{ or } 2. \quad \checkmark$$



$$\text{Let } \angle PAB = \theta, \angle CFD = z, \angle CQD = \beta$$

Now $\angle PAB = \angle PCA = \theta$ ✓

($\angle$ between tangent and chord at pt of contact = $\angle$ in alt. segment)

similarly $\angle PAB = \angle BDA = \theta$

$\therefore \angle FCQ = 180^\circ - \theta = \angle FDQ$

$\therefore$ In quad. FCQD :

$2 + 180^\circ - \theta + \theta + 180^\circ - \theta = 360^\circ$ ✓
($\angle$ sum of quad. = 360°)

$\therefore 2 + \theta = 2\theta$

$\therefore \theta = \frac{1}{2}(2 + \theta)$

i.e. $\angle PAB = \frac{1}{2}(\angle CFD + \angle CQD)$ ✓

(d) $I = 2 \int_0^{\frac{\pi}{4}} \cos^2 4x \, dx$

Now $\cos 2x = 2\cos^2 x - 1$

$\therefore \cos 8x = 2\cos^2 4x - 1$

$\therefore 2\cos^2 4x = 1 + \cos 8x$ ✓

$\therefore I = \int_0^{\frac{\pi}{4}} 1 + \cos 8x \, dx$

$= \left[x + \frac{\sin 8x}{8} \right]_0^{\frac{\pi}{4}}$ ✓

$= \left[\left(\frac{\pi}{4} + \frac{\sin 2\pi}{8} \right) - (0 + 0) \right]$

$= \frac{\pi}{4}$ ✓

(e) $\cos 5\theta - \cos 2\theta = 0$

$\therefore \cos 5\theta = \cos 2\theta$

$\therefore 5\theta = 2\pi n \pm 2\theta$, where n is any integer ✓

$\therefore 3\theta = 2\pi n$ or $7\theta = 2\pi n$

$\therefore \theta = \frac{2\pi n}{3}$ or $\frac{2\pi n}{7}$ ✓

3. (a) $y = x^2 - 4x$

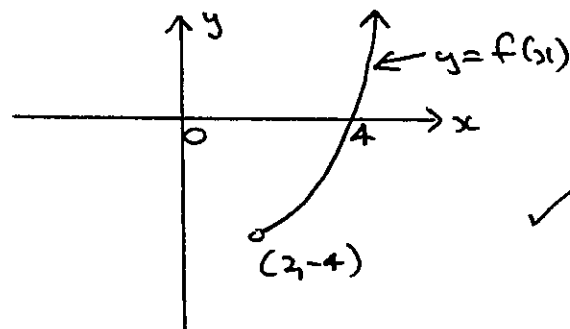
$\therefore \frac{dy}{dx} = 2x - 4$

For monotonic increasing $\frac{dy}{dx} > 0$

$\therefore x > 2$

(i) When $x = 2$ $y = -4$

$\therefore$ graph of $y = f(x)$ is:



(ii) For inverse function interchange x for y $\therefore x = y^2 - 4y$

$\therefore x = (y-2)^2 - 4$ ✓

$\therefore y-2 = \pm \sqrt{x+4}$

$\therefore y = 2 \pm \sqrt{x+4}$

$\therefore f^{-1}(x) = 2 + \sqrt{x+4}$ ✓

(iii) For inverse function:

Domain is: $x > -4$

Range is: $y > 2$. ✓

(b) (i) $f(x) = 3 \sin 2x - x$

Now $f(1.33) = 0.0595... > 0$

$f(1.34) = -0.0038... < 0$

$\therefore$ As $f(1.33)$ and $f(1.34)$ have opposite signs and $f(x)$ is cts $\forall x$

$\Rightarrow f(x)$ has at least 1 root in the interval $1.33 < x < 1.34$ ✓

(ii) $f(x) = 3 \sin 2x - x$

$\therefore f'(x) = 6 \cos 2x - 1$

By Newton's Method $z_2 = z_1 - \frac{P(z_1)}{P'(z_1)}$

$$\therefore \text{if } z_1 = 1.33$$

$$\therefore z_2 = 1.33 - \frac{P(1.33)}{P'(1.33)} \quad \checkmark$$

$$= 1.33 - \frac{0.0595...}{-6.3175...} \quad \checkmark$$

$$= 1.3394298...$$

$$= 1.3394 \text{ (4 d.p.)} \quad \checkmark$$

2) $y = \frac{x^2}{4-x^2}$

(i) For vertical asymptotes $4-x^2=0$

$$\therefore x = \pm 2$$

are vertical asymptotes at $x = \pm 2$

For horiz. asymptote $\lim_{x \rightarrow \pm\infty} y$

$$= \lim_{x \rightarrow \pm\infty} \frac{x^2(1)}{x^2\left(\frac{4}{x^2}-1\right)}$$

$$= \frac{1}{0-1} \left(\text{As } x \rightarrow \pm\infty, \frac{4}{x^2} \rightarrow 0 \right)$$

$$= -1$$

$\therefore$ horizontal asymptote at $y = -1$ $\checkmark$

(ii) $y = \frac{x^2}{4-x^2}$

$$\therefore \frac{dy}{dx} = \frac{(4-x^2) \cdot 2x - x^2(-2x)}{(4-x^2)^2}$$

$$= \frac{8x - 2x^3 + 2x^3}{(4-x^2)^2}$$

$$= \frac{8x}{(4-x^2)^2} \quad \checkmark$$

For a stat. pt $\frac{dy}{dx} = 0 \therefore x = 0$

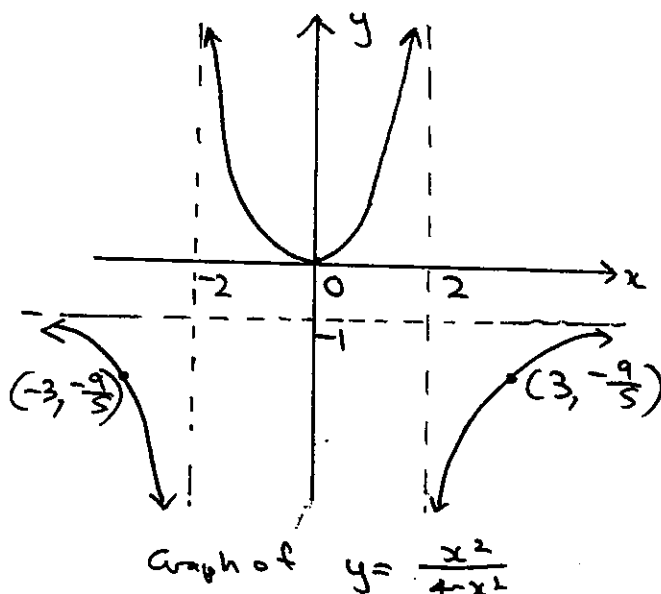
when $x=0$

x	0^-	0	0^+
y'	$-$	0	$+$

$\checkmark \Rightarrow$ min. turn pt at $(0,0)$ $\checkmark$

(iii) Let $y = f(x)$

$\therefore f(-x) = f(x) \therefore$ function is even.



4 (a) TO PROVE: $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n(n+1)^2$ for $n \geq 1$.

PROOF: Step 1: When $n=1$ LHS $= 1^3 = 1$
RHS $= \frac{1}{4} \cdot 1^2(2)^2 = 1$
 $=$ LHS

$\therefore$ it is true for $n=1$. $\checkmark$

Step 2: Assume it is true for $n=k$ ($k \in \mathbb{N}$, $n \in \mathbb{N}^+$) and prove it is true for $n=k+1$.

$$\text{Now } S_k + T_{k+1} = S_{k+1}$$

$$\therefore \text{LHS} = S_k + T_{k+1}$$

$$= \frac{1}{4}k^2(k+1)^2 + (k+1)^3$$

$$= (k+1)^2 \left[\frac{1}{4}k^2 + k+1 \right]$$

$$= (k+1)^2 \left[\frac{k^2 + 4k + 4}{4} \right] \quad \checkmark$$

$$= \frac{1}{4}(k+1)^2(k+2)^2$$

$$\text{RHS} = S_{k+1}$$

$$= \frac{1}{4}(k+1)^2(k+1+1)^2$$

$$= \frac{1}{4}(k+1)^2(k+2)^2$$

$$= \text{LHS.} \quad \checkmark$$

$\therefore$ if it is true for $n=k$ so it is true for $n=k+1$. $\checkmark$

Step 3: It is true for $n=1$ and so, it is true for $n=1+1=2$. It is true for $n=2$ and so, it is true for $n=2+1=3$ and so on for all positive integral values of n . ✓

(ii) Now $2^3 + 4^3 + 6^3 + \dots + 20^3$

$$= (2^3 \cdot 1^3) + (2^3 \cdot 2^3) + (2^3 \cdot 3^3) + \dots + (2^3 \cdot 10^3)$$

$$= 2^3 [1^3 + 2^3 + 3^3 + \dots + 10^3]$$

$$= 8 \left[\frac{1}{4} \times 10^2 \times 11^2 \right]$$

$$= 24200 \quad \checkmark$$

(b) $P(x) = 2x^3 - 5x^2 - 3x + 1$ has zeros α, β and γ .

$$\therefore \alpha + \beta + \gamma = -\frac{b}{a} = \frac{5}{2}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = -\frac{3}{2} \quad \checkmark$$

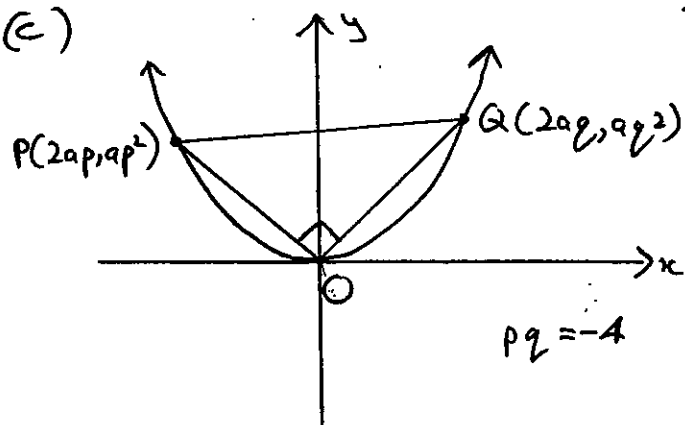
$$\alpha\beta\gamma = -\frac{d}{a} = -\frac{1}{2}$$

$$(i) 3\alpha + 3\beta + 3\gamma - 4\alpha\beta\gamma = 3\left(\frac{5}{2}\right) - 4\left(-\frac{1}{2}\right) = 9\frac{1}{2} \quad \checkmark$$

$$\begin{aligned} (ii) \alpha^{-1} + \beta^{-1} + \gamma^{-1} &= \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} \\ &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\ &= \frac{-3/2}{-1/2} \\ &= 3 \quad \checkmark \end{aligned}$$

$$\begin{aligned} (iii) \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= \left(\frac{5}{2}\right)^2 - 2\left(-\frac{3}{2}\right) \\ &= 9\frac{1}{2} \quad \checkmark \end{aligned}$$

(c)



$$M_{PQ} = \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right)$$

$$= (a(p+q), \frac{a(p^2+q^2)}{2}) \quad \checkmark$$

$$\text{Now } x = a(p+q) \therefore p+q = \frac{x}{a} \quad \text{--- (1)}$$

$$y = a\left(\frac{p^2+q^2}{2}\right) \therefore p^2+q^2 = \frac{2y}{a} \quad \text{--- (2)}$$

$$\text{Now } p^2+q^2 = (p+q)^2 - 2pq \quad \checkmark$$

$$\therefore \frac{2y}{a} = \left(\frac{x}{a}\right)^2 - 2(-4)$$

$$\therefore \frac{2y}{a} = \frac{x^2}{a^2} + 8$$

$$\therefore 2ay = x^2 + 8a^2$$

$$\therefore x^2 = 2ay - 8a^2$$

$$\therefore x^2 = 2a(y - 4a) \quad \checkmark$$

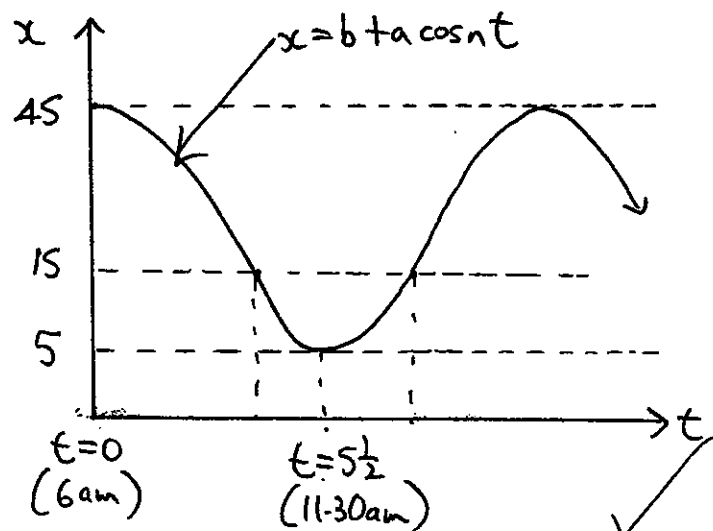
$\Rightarrow$ locus of the midpoint of PQ is a parabola with vertex $(0, 4a)$.

$$5(a) R = \left(\frac{kx_2 + lx_1}{k+l}, \frac{ky_2 + ly_1}{k+l} \right) \quad \checkmark$$

$$\therefore R = \left(\frac{5(5) + 6(2)}{5+6}, \frac{5(4) + 6(3)}{5+6} \right)$$

$$= \left(\frac{37}{11}, \frac{38}{11} \right) \quad \checkmark$$

(i)



$$\begin{aligned} \text{Now } 45 &= b + a \quad \text{--- (1)} \\ 5 &= b - a \quad \text{--- (2)} \end{aligned}$$

$$\text{(1) + (2): } 50 = 2b \therefore b = 25 \therefore a = 20$$

$$\text{Now period, } T = 2 \times 5\frac{1}{2} = 11 = \frac{2\pi}{n}$$

$$\therefore n = \frac{2\pi}{11}$$

$$\therefore x = 25 + 20 \cos \frac{2\pi}{11} t$$

$$\text{If } x = 15, 15 = 25 + 20 \cos \frac{2\pi}{11} t$$

$$\therefore -\frac{1}{2} = \cos \frac{2\pi}{11} t$$

$$\therefore \frac{2\pi}{11} t = \cos^{-1}\left(-\frac{1}{2}\right) \quad (\text{require 2nd, 3rd quadrants})$$

$$\therefore \frac{2\pi}{11} t = \pi - \frac{\pi}{3} \text{ or } \pi + \frac{\pi}{3}$$

$$\therefore t = \frac{2}{3} \times \frac{11}{2} \text{ or } \frac{4}{3} \times \frac{11}{2}$$

$$\therefore t = \frac{11}{3} \text{ or } \frac{22}{3}$$

$\therefore$ For Fls Flutos to release pigeons

for training earliest time interval

is: 6am + 3 hr 40mins to 6am
+ 7 hr 20mins

= 9.40 am to 1.20 pm.

$$b \text{ (i) Now } \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\therefore \text{ let } B = B+C$$

$$\therefore \tan(A+B+C) = \frac{\tan A + \tan(B+C)}{1 - \tan A \tan(B+C)}$$

$$= \frac{\tan A + \frac{\tan B + \tan C}{1 - \tan B \tan C}}{1 - \tan A \left(\frac{\tan B + \tan C}{1 - \tan B \tan C} \right)}$$

$$= \frac{\tan A(1 - \tan B \tan C) + \tan B + \tan C}{1 - \tan B \tan C - \tan A(\tan B + \tan C)}$$

$$= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

(*)

(ii) If A, B and C are $\angle$ s of a Δ

$$\therefore A+B+C = \pi \therefore \tan(A+B+C) = 0$$

$$\text{and as } \frac{\tan A}{5} = \frac{\tan B}{6} = \frac{\tan C}{7} = k$$

$$\therefore \tan A = 5k, \tan B = 6k, \tan C = 7k$$

$$\Rightarrow 0 = \frac{5k + 6k + 7k - (5k)(6k)(7k)}{1 - (5k)(6k) - (6k)(7k) - (7k)(5k)}$$

sub into (*)

$$\therefore 18k - 210k^3 = 0$$

$$\therefore 6k(3 - 35k^2) = 0$$

$$\therefore k = 0 \text{ or } \pm \sqrt{\frac{3}{35}}$$

But as $\tan A, \tan B, \tan C > 0 \therefore k > 0$

$$\therefore k = \sqrt{\frac{3}{35}} \text{ only.}$$

(iii) Now smallest angle is A

$$\text{where } \tan A = 5k$$

$$\therefore \tan A = 5\sqrt{\frac{3}{35}}$$

$$\therefore \angle A = 55^\circ 40' \quad (\text{to nearest min})$$

b) (i) Let $P(h) = h^3 - 9h^2 + 8$

if $h=1$ $P(1) = 1 - 9 + 8 = 0$

$\therefore h=1$ is a root. ✓

$\therefore P(h) = (h-1)(h^2 - 8h - 8)$

$\therefore$ other roots are solved from $h^2 - 8h - 8 = 0$

$\therefore h = \frac{8 \pm \sqrt{64 - 4 \cdot 1 \cdot -8}}{2}$

$= \frac{8 \pm \sqrt{96}}{2}$

$= \frac{8 \pm 4\sqrt{6}}{2}$

$= 4 \pm 2\sqrt{6}$ ✓

$\therefore$ roots are $h=1, 4 \pm 2\sqrt{6}$ ✓

(ii) $V = \frac{\pi}{3}(9h^2 - h^3)$, $\frac{dV}{dt} = \frac{\pi}{3} \text{ m/min}$

(2) when $t=8$ $V = 8\left(\frac{\pi}{3}\right) = \frac{8\pi}{3} \text{ m}^3$

$\therefore \frac{8\pi}{3} = \frac{\pi}{3}(9h^2 - h^3)$

$\therefore 8 = 9h^2 - h^3$

$\therefore h^3 - 9h^2 + 8 = 0$ ✓

$\therefore h = 1, 4 \pm 2\sqrt{6}$ (from (i) above).

But $0 < h \leq 3$ (radius of bowl is 3m)

$\therefore$ if $h = 4 + 2\sqrt{6}$ this is greater than 3

and if $h = 4 - 2\sqrt{6}$ " " less than 0.

$\therefore$ depth after 8 mins is 1m. ✓

(B) $\frac{dV}{dh} = \frac{\pi}{3}(18h - 3h^2)$

$\therefore$ when $h=1$ $\frac{dV}{dh} = \frac{\pi}{3}(15) = 5\pi$

and $\frac{dV}{dt} = \frac{\pi}{3}$ ✓

Now. $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ ✓

$\therefore$ when $h=1$ $\frac{\pi}{3} = 5\pi \cdot \frac{dh}{dt}$

$\therefore \frac{dh}{dt} = \frac{1}{15} \text{ m/min.}$

$\therefore h$ is increasing at $\frac{1}{15} \text{ m/min.}$ ✓

(C) $\ddot{x} = \frac{-4}{(x+2)^2} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$ ✓

$\therefore \frac{1}{2}v^2 = -4 \int (x+2)^{-2} dx$
 $= \frac{4}{x+2} + c$

when $x=-1, v=6 \therefore 18 = 4 + c$

$\therefore c = 14$ ✓

$\therefore \frac{1}{2}v^2 = \frac{4}{x+2} + 14$

$\therefore v^2 = \frac{8}{x+2} + 28$

$\therefore v = \sqrt{\frac{8}{x+2} + 28}$ ✓

(taking the square root as initial condition gave a true velocity)

when $x=6$ $v = \sqrt{1 + 28} = \sqrt{29}$

$\therefore$ velocity of π -meson when it is 6m to the right of O is $\sqrt{29} \text{ ms}^{-1} \rightarrow$ ✓

6 (a) (i) $\ddot{x} = -n^2(x-b)$ — (1)

$x = b + a \cos nt$ — (2)

sub (2) into (1):

LHS of (1) = $\ddot{x}$

$= \frac{d}{dt}(\dot{x})$

$= \frac{d}{dt}(-an \sin nt)$

$= -an^2 \cos nt$

$= -n^2(x-b)$ [as $a \cos nt = x-b$ from (2)]

$= \text{RHS of (1)}$

$\therefore x = b + a \cos nt$ satisfies the given equation. ✓

$$1) (a) (i) \frac{dT}{dt} = -k(T-20) \quad \text{--- (1)}$$

$$T = 20 + Ae^{-kt} \quad \text{--- (2)}$$

sub (2) into (1):

$$\begin{aligned} \text{LHS of (1)} &= \frac{dT}{dt} \\ &= \frac{d}{dt}(20 + Ae^{-kt}) \\ &= -kAe^{-kt} \\ &= -k(T-20) \quad \left(\begin{array}{l} \text{As } Ae^{-kt} \\ = T-20 \\ \text{from (2)} \end{array} \right) \\ &= \text{RHS of (1)} \end{aligned}$$

$\Rightarrow T = 20 + Ae^{-kt}$ satisfies given equation.

$$(ii) T = 20 + Ae^{-kt}$$

$$\text{when } t=0, T=95$$

$$\therefore 95 = 20 + Ae^0 \therefore A = 75$$

$$\therefore T = 20 + 75e^{-kt}$$

$$\text{when } t=10, T=70$$

$$\therefore 70 = 20 + 75e^{-10k}$$

$$\therefore \frac{50}{75} = e^{-10k}$$

$$\therefore k = -\frac{1}{10} \ln \frac{2}{3}$$

$$\therefore T = 20 + 75e^{(\frac{1}{10} \ln \frac{2}{3})t}$$

$$\text{when } t=15 \quad T = 20 + 75e^{(\frac{1}{10} \ln \frac{2}{3})15}$$

$$\therefore T = 20 + 75 \left(\frac{2}{3} \right)^{3/2}$$

$$\therefore T = 60.824 \dots$$

$\Rightarrow$ temp. of soup is then 61°C
(to nearest degree)

$$(iii) \text{ when } T=35, t=?$$

$$\therefore 35 = 20 + 75e^{(\frac{1}{10} \ln \frac{2}{3})t}$$

$$\therefore \frac{1}{5} = e^{(\frac{1}{10} \ln \frac{2}{3})t}$$

$$\therefore t = \frac{\ln \frac{1}{5}}{\frac{1}{10} \ln \frac{2}{3}}$$

$$\therefore t = 39.693 \dots$$

$\therefore$ it will take 40 mins (to nearest min)

$$(iv) \frac{dT}{dt} = \left(\frac{1}{10} \ln \frac{2}{3} \right) (T-20) \quad (\text{from (1)})$$

$$\text{when } T=35 \quad \frac{dT}{dt} = -0.60819 \dots$$

$\therefore$ rate of cooling is -0.6°C/min
(to 1 d.p.)

$$(b) \sin x - 7 \cos x = -5, 0 \leq x \leq 2\pi$$

$$\text{Now } R = \sqrt{1^2 + (-7)^2} = 5\sqrt{2}$$

$$\therefore 5\sqrt{2} \left(\frac{1}{5\sqrt{2}} \sin x - \frac{7}{5\sqrt{2}} \cos x \right) = -5$$

$$\therefore 5\sqrt{2} \sin(x-\alpha) = -5$$

$$\text{where } \cos \alpha = \frac{1}{5\sqrt{2}}, \sin \alpha = \frac{7}{5\sqrt{2}}$$

$$\therefore \tan \alpha = 7 \therefore \alpha = \tan^{-1} 7$$

$$\therefore \sin(x - \tan^{-1} 7) = -\frac{1}{\sqrt{2}}$$

$$\therefore \text{Basic angle } x - \tan^{-1} 7 = \frac{\pi}{4} \quad (\text{requires 3rd \& 4th quads})$$

$$\therefore x = \pi + \frac{\pi}{4} + \tan^{-1} 7 \quad \text{or} \quad 2\pi - \frac{\pi}{4} + \tan^{-1} 7$$

$$\therefore x = 5.36 \quad (2 \text{ d.p.})$$

$$\text{or } x = 0.64$$

(c) $y = \tan^{-1}(\sin 3x)$, $0 \leq x \leq \pi$.

$$\frac{dy}{dx} = \frac{1}{1 + (\sin 3x)^2} \cdot 3 \cos 3x$$

For a stat. pt $\frac{dy}{dx} = 0$

$$\therefore 3 \cos 3x = 0$$

$$\therefore 3x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \quad \text{for } 0 \leq 3x \leq 3\pi$$

$$\therefore x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6} \quad \text{for } 0 \leq x \leq \pi$$

when $x = \frac{\pi}{6}$

x	$\frac{\pi}{6}^-$	$\frac{\pi}{6}$	$\frac{\pi}{6}^+$
y'	+	0	-

$\Rightarrow$ max turn pt at $(\frac{\pi}{6}, \frac{\pi}{4})$

when $x = \frac{\pi}{2}$

x	$\frac{\pi}{2}^-$	$\frac{\pi}{2}$	$\frac{\pi}{2}^+$
y'	-	0	+

$\Rightarrow$ min. turn pt at $(\frac{\pi}{2}, -\frac{\pi}{4})$

when $x = \frac{5\pi}{6}$

x	$\frac{5\pi}{6}^-$	$\frac{5\pi}{6}$	$\frac{5\pi}{6}^+$
y'	+	0	-

$\Rightarrow$ max turn pt at $(\frac{5\pi}{6}, \frac{\pi}{4})$

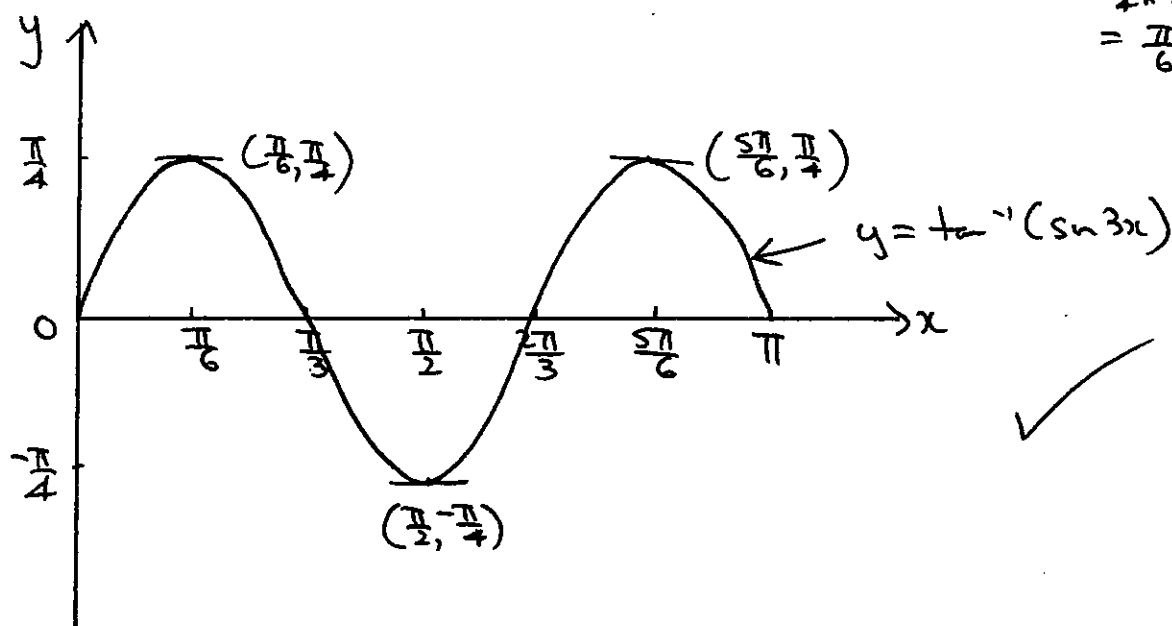
when $y = 0$

$$\sin 3x = 0$$

$$\therefore 3x = 0, \pi, 2\pi, 3\pi$$

$$\therefore x = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi$$

$$\text{period} = \frac{2\pi}{3} \therefore \text{sub-interval width} = \frac{1}{2} \times \frac{2\pi}{3} = \frac{\pi}{6}$$



Markers Comments Cranbrook trial 2009

Q1 a) Working should be shown for these questions. Extension! Mathematics

b) LOG LAWS too many students struggled with chain and quotient rules - apart from not getting it right time is not allowed for this in this type of question

c) Well done by many - easy marks for learning basic rules!

d) as above giveaway marks for those who know the pattern

e) too many assumed $x^2 = \cos^2 \theta$!? instead of the ol' $\cos^2 \theta + \sin^2 \theta = 1$

f) not well done - always simplify ASAP
Another way: $\sqrt{x}^2 = u^2 \therefore x = u^2 \quad \frac{dx}{du} = 2u$
(squaring) $\therefore dx = 2u du$

- 2a) Very good
- b) Very good - a few left as factors $(x+1)$ etc... = 0 instead of finding roots $x=1$..
- c) Too few tried - it's important to attempt all questions writing down a few obvious theorems earn't most who did so at least 1 mark
NEVER LEAVE A BLANK!
- d) Well done & again easy 'book work' marks
- e) The basic pattern! so simple & so quick ☺
always read questions a couple of times
and look for basic patterns ☺

Markers Notes

Math Ext 1 2009 Cranbrook - JSH

Q3 and Q4: Coordinates of a point in the number plane requires BRACKETS! Eg. The point (2, 3) (and not just 2, 3!)

Q3a. Some people didn't leave an open point at the left of the domain. Most found the inverse function okay. Some only interchanged x, y but failed to find the inverse function explicitly as a function of x .

Q3b Done well.

Q3ci Done well

Q3cii. Done well overall.

Q3ciii. Mostly done well.

Q4a. Many people misunderstood the induction statement. The case when $n = k$ says that $1^3 + 2^3 + \dots + k^3 = k^2(k+1)^2/4$ (NOT $k^3 = k^2(k+1)^2/4$)

$$\text{Q4b. } \alpha^{-1} + \beta^{-1} + \gamma^{-1} = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$
$$\left(\text{not } \frac{1}{\alpha+\beta+\gamma}! \right)$$

Q4c. Several people forgot the mid-point formula!!!!

A few people decided to include a random calculation deriving the equation of the chord - NOT ASKED FOR - $\therefore$ marks=0 $\therefore$ = waste of time.

5 (a) Mostly well done but a few students either could not remember the division formula or could not substitute into it properly.

(b) (i) Some students had problems with establishing that $h^3 - 9h^2 + 8 = (h-1)(h^2 - 8h + 8)$

(2) (ii) Most students did not justify why $h=1$ was the only possible value for h .

(B) (ii) Some students did not use $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ to find $\frac{dh}{dt}$.

(c) If acceleration ($\ddot{x}$) is in terms of x then in order to find v we must use $\ddot{x} = \frac{d}{dx}(\frac{1}{2}v^2)$. Some students did not realise this was could not proceed successfully with this question

6 (a) (i) Mostly well done but some students forgot to justify why $\arcsin$ was $x-b$ to obtain full marks.

(ii) Generally most students made a good attempt at this question.

(b) (i) Letting $B = B+C$ gave the result from the $\tan(A+B)$ expansion. Not well done here.

(ii) Few students could follow on from part (i) not realising that $\tan(A+B+C) = 0$.

(iii) Generally well done.

7. (a) (i) Well done.

(ii) ~~part~~ generally good;

(iii) ^{2nd part:} Some students did not read the question correctly and substituted $t=5$ instead of $t=15$ for 'a further 5 minutes'.

(iii) Mostly well done.

(iv) Some substitution problems

(b) As the domain was $0 \leq x \leq 2\pi$ a solution of $x = 6.93$ (2dp) was ~~a~~ outside this domain and 2π was needed to be subtracted from this to obtain 0.64 as the 2nd correct solution.

(c) Some students just drew the graph ignoring the requirements to find the stationary points and determining their nature.